GreenLink Delhi

A budget-aware geospatial decision-support tool for planning urban green corridors

Project Description

GreenLink Delhi is a geospatial planning prototype that helps urban planners explore how green spaces could be connected through potential green corridors when funding is limited. It combines mapped green-space features, a street-network view, corridor candidates and a budget-constrained selection approach to make planning trade-offs easier to understand. Users can compare strategies that prioritize ecological connectivity or a heat-priority spatial proxy, inspect estimated costs, and see how a chosen set of candidate corridors performs against the selected objective. The tool is intended to support early-stage planning and discussion—not replace field surveys, engineering design, ecological assessment or official approval.

Core idea: make urban greening decisions more transparent by showing where investment could go, what it may cost, and how different priorities change the recommended plan.

The Problem

Urban green spaces can be disconnected from one another, while city agencies and communities face finite budgets and competing priorities. Choosing where to invest is difficult when connectivity, heat-related priorities, route feasibility and cost must be considered together. A map alone shows where green spaces are; it does not explain which connections might be worth prioritizing under a fixed budget.

The Proposed Solution

- Map green-space features and candidate links in a defined South-Central Delhi study area.
- Estimate the cost of each candidate corridor using a provisional, typology-based cost model.
- Score candidates against a selected planning objective, such as improving connectivity or prioritizing areas represented by a heat-related proxy.
- Select a set of candidates under a user-specified budget using a greedy marginal-benefit heuristic.
- Compare scenarios so decision-makers can see how changing the objective changes the selected routes, spending and reported metrics.

How It Works

The workflow begins with mapped green-space features and street-network data. The prototype represents possible corridor connections, estimates their costs and evaluates their contribution to the selected objective. A budget-constrained heuristic then builds a candidate portfolio. The resulting scenarios can be compared using cost, connectivity indicators and the heat-priority proxy. The output is a shortlist for further investigation rather than a construction-ready design.

Current Prototype at a Glance

Study area	South-Central Delhi
Mapped green-space features	12 OSM-derived polygons
Candidate corridors	14
Total candidate length	Approximately 46.95 km

Estimated cost across candidates	Approximately ■17.16 crore

These figures describe the current prototype dataset and should be checked against the live app before presentation. OSM-derived features may be incomplete or unverified on the ground.

Scenario Comparison and Illustrative Findings

The prototype compares two planning profiles under the same illustrative ■5 crore budget:

- **Connectivity-first:** prioritizes links that improve the prototype's distance-decay connectivity indicator. The reported scenario spends about ■4.98 crore and increases the indicator from 15.17 to 17.35.
- **Heat-priority-first:** prioritizes corridor overlap with a weighted spatial proxy grid. The reported scenario spends about ■4.80 crore and records 40.48 weighted hectares for the proxy, with a secondary connectivity indicator of 16.42.
- **Trade-off:** the scenarios share some candidate corridors but differ elsewhere, illustrating that the recommended portfolio depends on the objective—not simply on selecting the shortest or cheapest route.

The heat-priority layer is a spatial proxy; it is not a measurement of land-surface temperature and does not prove that a corridor will deliver a particular cooling effect.

Technology and Methods

- **Frontend:** React, TypeScript, Vite, Tailwind CSS and MapLibre GL JS for the interactive map interface.
- **Backend:** FastAPI for serving planning and optimization functionality.
- **Spatial inputs:** OpenStreetMap-derived features and geospatial data such as GeoJSON.
- **Planning logic:** street-network routing and a greedy marginal-benefit heuristic for budget-constrained candidate selection.
- **Evaluation:** scenario comparisons using estimated spend, connectivity indicators and a weighted heat-priority proxy.

Potential Value

GreenLink Delhi can help urban-planning teams, sustainability groups, researchers and civic innovators compare early-stage greening strategies before committing resources to detailed studies. Its main value is explainability: it connects map-based exploration with explicit budgets, objectives, cost estimates and measurable scenario outputs.

Limitations and Responsible Use

- Mapped green spaces and street data come from OpenStreetMap-derived sources and require ground verification.
- Costs are provisional estimates based on road typology, not contractor quotations or approved budgets.
- The heat-priority measure is a proxy based on spatial overlap, not satellite-derived land-surface temperature or a validated cooling model.
- The greedy heuristic is a practical selection method and is not guaranteed to find the globally optimal portfolio.
- Candidate corridors are planning concepts, not approved planting designs; ownership, utilities, accessibility, biodiversity, maintenance and community needs require further assessment.

Future Scope

- Allow users to choose locations or green-space endpoints and generate route candidates on demand, alongside the existing precomputed-corridor demo mode.

- Add stronger feasibility checks for road width, ownership, utilities, pedestrian access and planting suitability.
- Integrate validated temperature and canopy datasets where licensing and data quality permit.
- Support field verification, stakeholder feedback, sensitivity analysis and more advanced optimization methods.
- Add verified high-detail basemaps or street-level imagery only where provider coverage, API access, licensing and attribution requirements are satisfied.

Short Project Pitch

GreenLink Delhi turns urban green-corridor planning into a transparent, budget-aware decision process. By mapping green spaces, evaluating candidate connections and comparing different planning priorities, it helps users understand how a limited budget can be allocated—and what trade-offs come with each strategy. It is a decision-support prototype for better-informed planning, with ground validation and feasibility studies remaining essential before implementation.